Determinants

T.T.

Q1

Apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

These row additions do not change the determinant, and they give

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & -1 & -2 \end{bmatrix}.$$

Now apply $R_3 \leftarrow R_3 + R_2$, again without changing the determinant:

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}.$$

There were no row exchanges or row scalings. The pivots are 1, 1, and -1, so

$$\det A = \det U = 1 \cdot 1 \cdot (-1) = -1.$$

Q2

(i) Write the rows of A as $r_1, \ldots, r_n$. The rows of cA are $cr_1, \ldots, cr_n$. Linearity in the first row gives one factor of c, and applying the same rule to each remaining row gives

$$\begin{aligned} \det(cA) &= \det(cr_1, \ldots, cr_n) \\ &= c^n \det(r_1, \ldots, r_n) \\ &= c^n \det A. \end{aligned}$$

(ii) Since A is 4×4,

$$\det(2A) = 2^4 \det A = 16(-2) = -32.$$

The product and transpose rules give

$$\det(A^T A) = \det(A^T) \det A = (\det A)^2 = 4.$$

Finally,

$$\det(A^{-1}) = \frac{1}{\det A} = -\frac{1}{2}.$$

(iii) Take $A = B = I_2$. Then

$$\det(A + B) = \det(2I_2) = 4,$$

whereas

$$\det A + \det B = 1 + 1 = 2.$$

Therefore the determinant is not linear in the whole matrix. Its linearity applies to one row or one column while all the others remain fixed.

Q3

(i) If the entries in each row add to zero, then multiplying A by the all-ones vector adds the entries in each row. Hence

$$A\mathbf{1} = 0.$$

The vector $\mathbf{1}$ is nonzero, so A has a nontrivial null vector. Therefore A is singular and

$$\det A = 0.$$

(ii) If the entries in each row add to one, then

$$A\mathbf{1} = \mathbf{1}.$$

It follows that

$$(A - I)\mathbf{1} = 0.$$

Thus $A - I$ is singular, and consequently

$$\det(A - I) = 0.$$

(iii) The condition does not imply $\det A = 1$. For example,

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

has row sums equal to one, but $\det A = 0$. The condition proves only that $A\mathbf{1} = \mathbf{1}$, or equivalently that $A - I$ has the nonzero null vector $\mathbf{1}$.

Q4

(i) Suppose the columns are dependent. Then there are scalars $c_1, \ldots, c_n$, not all zero, such that

$$c_1 a_1 + \cdots + c_n a_n = 0.$$

Choose k with $c_k \neq 0$. Solving for the kth column gives

$$a_k = -\sum_{j \neq k} \frac{c_j}{c_k} a_j.$$

Use linearity in column k while leaving every other column fixed. Then $\det A$ becomes a linear combination of determinants in which column k equals some column $j \neq k$. Every such determinant has two equal columns and is therefore zero. Hence

$$\det A = 0.$$

(ii) Conversely, suppose $\det A = 0$. Then A is not invertible. A square matrix is invertible exactly when its columns are linearly independent, so the columns of A must be dependent. Equivalently, there is a nonzero vector x such that

$$Ax = x_1 a_1 + \cdots + x_n a_n = 0.$$

(iii) The columns of A are the edge vectors of the parallelepiped obtained from the unit cube by multiplication by A. If the columns are dependent, they lie in a lower-dimensional subspace, so the parallelepiped is flattened and has zero n-dimensional volume. This agrees with

$$\mathrm{Vol} = |\det A| = 0.$$

Q5

(i) Taking determinants of $Q^TQ = I$ gives

$$\det(Q^T)\det Q = \det I = 1.$$

Since $\det(Q^T) = \det Q$,

$$(\det Q)^2 = 1.$$

The determinant is real, so

$$\det Q = 1 \qquad \text{or} \qquad \det Q = -1.$$

(ii) Let the first column be

$$q_1 = (\cos\theta, \sin\theta)^T.$$

The second column must be a unit vector perpendicular to q_1. The perpendicular unit vectors are

$$q_2 = (-\sin\theta, \cos\theta)^T \quad \text{and} \quad q_2 = (\sin\theta, -\cos\theta)^T.$$

In the first case,

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \det Q = \cos^2\theta + \sin^2\theta = 1.$$

In the second case,

$$Q = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}, \qquad \det Q = -\cos^2\theta - \sin^2\theta = -1.$$

The first matrix preserves orientation, while the second reverses it.

Q6

(i) The cofactor formula is

$$A^{-1} = \frac{1}{\det A}C^T.$$

Because A has integer entries, every minor and every cofactor is an integer. If $\det A = 1$ or $\det A = -1$, division by $\det A$ changes only the signs of the cofactors. Therefore every entry of A^{-1} is an integer.

(ii) If A and A^{-1} both have integer entries, then their determinants are integers. Since

$$\det A \;\det(A^{-1}) = \det I = 1,$$

two integers multiply to 1. The only possibilities are

$$\det A = 1, \quad \det(A^{-1}) = 1,$$

or

$$\det A = -1, \quad \det(A^{-1}) = -1.$$

Thus $\det A = 1$ or $\det A = -1$.

Q7

(i) Writing the matrix-vector product as a combination of columns gives

$$b = Ax = x_1 a_1 + \cdots + x_n a_n.$$

(ii) The determinant is linear in column j, with all other columns fixed. Therefore

$$\begin{aligned} \det A_j(b) &= \det A_j(x_1 a_1 + \cdots + x_n a_n) \\ &= \sum_{k=1}^{n} x_k \det A_j(a_k). \end{aligned}$$

If $k \neq j$, the matrix $A_j(a_k)$ has two identical columns: its original column k and its replaced column j. Hence

$$\det A_j(a_k) = 0 \qquad (k \neq j).$$

When $k = j$, the matrix $A_j(a_j)$ is the original matrix A. Only one term remains, so

$$\det A_j(b) = x_j \det A.$$

(iii) Since A is invertible, $\det A \neq 0$. Dividing the identity in part (ii) by $\det A$ gives

$$x_j = \frac{\det A_j(b)}{\det A}, \qquad j = 1, \ldots, n.$$

This is Cramer's rule.

Q8

The (i, j) entry of AC^T is

$$(AC^T)_{ij} = \sum_{k=1}^{n} a_{ik} C_{jk}.$$

(i) When $i = j$, this sum is the cofactor expansion of $\det A$ along row i:

$$(AC^T)_{ii} = \sum_{k=1}^{n} a_{ik} C_{ik} = \det A.$$

Thus every diagonal entry equals $\det A$.

(ii) Suppose $i \neq j$. Form a matrix B by replacing row j of A with row i. The cofactors of the new row j are still $C_{j1}, \ldots, C_{jn}$, because deleting row j removes the replaced row. Expanding $\det B$ along row j gives

$$\det B = \sum_{k=1}^{n} a_{ik} C_{jk} = (AC^T)_{ij}.$$

But B has two equal rows, namely rows i and j. Therefore $\det B = 0$, and every off-diagonal entry of AC^T is zero.

(iii) Combining the diagonal and off-diagonal results gives

$$AC^T = (\det A) I.$$

If $\det A \neq 0$, divide by $\det A$:

$$A \left(\frac{1}{\det A} C^T \right) = I.$$

Therefore

$$A^{-1} = \frac{1}{\det A} C^T.$$